

Agent Name: Geospatial Supply Chain Agent

What it does:

The Geospatial Supply Chain Intelligence Agent leverages a robust Supply Chain Knowledge Graph to deliver context-aware recommendations across suppliers, products and locations. By connecting and analyzing relationships between suppliers, products and locations the agent enables to make faster and smarter supply chain decisions. The agent helps to identify alternate suppliers based on product availability, pricing, and regional proximity. The most challenging part in the supply chain is to search the nearest location for alternative suppliers. This agent solves the issues using geospatial co-ordinates to find the suppliers.

Key capabilities include:

- Alternate supplier recommendation based on availability, location, and supplier relationships
- Nearest supplier discovery using geospatial graph queries
- Alternate suppliers for similar products

Dataset and why a graph fits:

Created a Supply Chain Dataset with 8000 rows having the below columns

Column	Description
Supplier_ID	Unique codes like SUP83421
Supplier_Name	Tech distributor names (e.g. Nova Components, Apex Distributors)
Product	48 distinct computer spare types (RAM, SSDs, CPUs, GPUs, laptop parts, peripherals, etc.)
Description	Detailed product specs per item
Price	pricing per product category
City	30 Major Indian cities (Mumbai, Bengaluru, Delhi, Hyderabad, Chennai, etc.)
Latitude	City coordinates
Longitude	City coordinates

A graph database is an ideal fit for supply chain systems because supply chains are naturally built around interconnected relationships between suppliers, products and locations. Supply Chains Are Highly Connected Networks

A single product may involve:

Multiple suppliers

Different transportation routes

Graph databases are designed specifically to manage and analyze these interconnected ecosystems efficiently.

Faster Alternate Supplier Discovery

When a supplier disruption occurs, graph queries can instantly identify:

Alternate suppliers

Nearby suppliers

Suppliers providing similar products

Suppliers connected through existing logistics routes

Instead of performing multiple complex table joins, the graph traverses relationships directly.

Geospatial and Nearest Supplier Intelligence

Graph databases can combine relationship intelligence with spatial data to:

Find the nearest supplier

Optimize delivery routes

Loaded the supply_chain_dataset.csv file from Github into Neo4j Aura using the below script:

LOAD CSV WITH HEADERS FROM <Github Path>

AS row

CALL(row) {

MERGE (sup:Supplier {supplierId: trim(row.Supplier_ID)}) SET sup.name=row.Supplier_Name

MERGE (prd:Product {name: trim(row.Product)}) SET

prd.description=row.Description,prd.price=toFloat(row.Price)

MERGE (cy:City {name: trim(row.City)}) SET cy.location = point({latitude: toFloat(row.Latitude), longitude: toFloat(row.Longitude)})

MERGE (sup)-[:SUPPLIES]->(prd)

MERGE (sup)-[:LOCATED_IN]->(cy)

} IN TRANSACTIONS OF 1000 ROWS

Agents in Aura console:

← → ↺ console.neo4j.io/projects/4c138448-ba4a-5a40-9

Create agent from scratch

Name

Supply Chain Assistant



Description

Helps find suppliers based on their location, products supplier

Prompt instruction

Answer questions about suppliers and products directly and accurately, providing relevant details from the database.

Instance

Development

Agent location ⓘ

GCP europe-west1

Access

☒ Internal

Free

Available to members of this Aura project

☐ External

\$0.35 / hour

Provided via an auto-generated endpoint

☐ Enable MCP server

← → ↺ console.neo4j.io/projects/4c138448-ba4a-5a40-9

Create agent from scratch

Available to members of this Aura project

☐ External

\$0.35 / hour

Provided via an auto-generated endpoint

☐ Enable MCP server

Agent tools

Add tool ▾

The agent will use its tools to perform actions on the instance

☒

Suppliers in City

Cypher Template





☒

Suppliers of Product

Cypher Template





☒

Alternate Suppliers for Product

Cypher Template





☒

Suppliers Nearest to City

Cypher Template





☒

Natural Language to Cypher Tool

Text2Cypher





☒

Alternate Suppliers for Similar Product

Cypher Template





Cancel

Update agent

Demo of Agents in Action:

Finding suppliers in a specific City:

MATCH (s:Supplier)-[:LOCATED_IN]->(c:City {name: 'Bengaluru'})

RETURN s.name as SupplierName,

s.supplierId as SupplierId,

c.name as City,

c.location as Location

ORDER BY s.name

Find suppliers in Bengaluru



Thought for 5 seconds

Here are the suppliers located in Bengaluru:

- Global Components
- Inno Depot
- Micro Hub
- Micro Mart
- Orbit Enterprises
- Prime Depot
- Rapid Mart
- Smart Electronics
- Smart Parts
- Smart Systems
- Star Zone

Ask anything

All information should be verified.

Finding the no of suppliers in a specific city and listing them

How many supplier are there in Hyderabad and list them



Thought for 6 seconds

There are 14 suppliers in Hyderabad. They are:

- Core Components
- Core Systems
- Cyber Distributors
- Cyber Hub
- Global Depot
- Global Distributors
- Indo Parts
- Inno Parts
- Micro Tech
- Orbit Mart
- Rapid Traders
- Smart Tech
- Star Tech
- Tech Works

Ask anything



Reasoning

Ah, I understand. The user wants to know how many suppliers are located specifically in Hyderabad. My immediate thought is to leverage the `Suppliers\_in\_City` tool. This tool is designed precisely for this kind of query, allowing me to fetch a comprehensive list of all suppliers associated with a particular city. Once I have that list, the final step is straightforward: I'll simply count the number of entries in the returned data to provide the exact quantity the user is seeking.



Applying agent tool: Suppliers in City

</> Input

```
{ "cityName": "Hyderabad" }
```

</> Output

```
{ "keys": [ "SupplierName", "SupplierId", "City", "Location" ], "records": [ { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP18326", "SupplierName": "Core Components" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP86484", "SupplierName": "Core Systems" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP48469", "SupplierName": "Cyber Distributors" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP99192", "SupplierName": "Cyber Hub" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP11504", "SupplierName": "Global Depot" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP16175", "SupplierName": "Global Distributors" }, { "City": "Hyderabad", "Location": { "SpatialRefId": 4326, "X": 78.523714, "Y": 17.400747 }, "SupplierId": "SUP38746", "SupplierName": "Indo
```

Finding the Suppliers of a Product & their location:

Who are the supplier of Laptop Battery 4400mAh and where are they located?



Thought for 12 seconds



Here are the suppliers of "Laptop Battery 4400mAh" and their locations:

- Zenith Electronics (Vadodara)
- Smart Systems (Bengaluru)
- Rapid Tech (Coimbatore)
- Nexus Enterprises (Visakhapatnam)
- Tech Zone (Thiruvananthapuram)
- Global Hub (Chandigarh)
- Inno Hub (Mumbai)
- Tech Parts (Visakhapatnam)
- Nexus Hub (Vadodara)
- Digital Enterprises (Ahmedabad)
- Cyber Parts (Kochi)
- Global Mart (Gurugram)
- Cyber Systems (Agra)
- Inno Systems (Nashik)
- Digital Distributors (Nashik)

Ask anything



Finding alternate nearest suppliers to a supplier which has closed in a specific location:

Nova Suppliers in Kochi is closed what are the alternate suppliers very close to them.



Thought for 6 seconds

Here are some alternate suppliers located in cities nearest to Kochi:

- **Coimbatore** (145 km away):
 - Apex Distributors
 - Rapid Tech
 - Global Tech
 - Apex Zone
 - Inno Zone
 - Nexus Enterprises
 - Zenith Parts
 - Cyber Mart
 - Sigma Mart
 - Swift Components
 - Indo Depot
 - Inno Solutions
 - Swift Electronics
 - Inno Distributors
- **Thiruvananthapuram** (176 km away):
 - Cyber Distributors
 - Tech Zone
 - Inno Systems
 - Nexus Hub
 - Core Systems

Ask anything

```
MATCH (c1:City {name: 'Kochi'})
MATCH (c2:City)
WHERE c1 <> c2
WITH c1, c2, point.distance(point({latitude: c1.latitude, longitude: c1.longitude}), point({latitude: c2.latitude, longitude: c2.longitude})) / 1000 AS distanceKm
ORDER BY distanceKm ASC
MATCH (s:Supplier)-[:LOCATED_IN]->(c2)
RETURN s.name AS supplierName, c2.name AS cityLocatedIn, distanceKm
```


What are Top 5 costly products supplied by a Supplier

What are the top 5 high costly unique product supplied by Swift Components?



Thought for 10 seconds



The top 5 most costly unique products supplied by Swift Components are:

- 1 GPU NVIDIA RTX 3060 - ₹35,882.65
- 2 CPU Intel Core i7-12700 - ₹30,529.77
- 3 Monitor 24" FHD - ₹13,779.94
- 4 CPU AMD Ryzen 5 5600 - ₹13,005.53
- 5 RAM 16GB DDR4 - ₹6,500.00

Which suppliers are very close to a specific location and minimum:

Which suppliers are near to Bhubaneswar and distance is also minimum.



Thought for 6 seconds

The suppliers nearest to Bhubaneswar are located in **Kolkata**, at a distance of 358 km.

Here are the suppliers in Kolkata:

- Tech Hub
- Inno Traders
- Star Parts
- Apex Hub
- Cyber Enterprises
- Nova Electronics
- Apex Traders
- Prime Works
- Core Depot
- Swift Parts

```
MATCH (targetCity:City {name: 'Bhubaneswar'})
```

```
WITH targetCity
```

```
MATCH (s:Supplier)-[:LOCATED_IN]->(c:City)
```

```
WHERE c <> targetCity
```

```
RETURN s.name AS supplierName, c.name AS cityLocatedIn,round(point.distance(c.location,
```

targetCity.location)/1000) AS distanceKm ORDER BY distanceKm

Thought for 6 seconds

Reasoning

My thought process begins with understanding the user’s request: they’re seeking suppliers located in proximity to Bhubaneswar, with a specific emphasis on finding those that are geographically closest. This immediately brings to mind a tool designed for exactly this purpose: ‘Suppliers_Nearest_to_City’. Therefore, my next step is to invoke this tool, making sure to specify “Bhubaneswar” as the ‘referenceCityName’ parameter. This ensures the tool will execute its function with the correct geographical context to identify the suppliers nearest to the requested city.

Applying agent tool: Suppliers Nearest to City

Input

```
{ "referenceCityName": "Bhubaneswar" }
```

Output

```
{ "keys": [ "supplierName", "cityLocatedIn", "distanceKm" ], "records": [ { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Tech Hub" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Inno Traders" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Star Parts" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Apex Hub" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Cyber Enterprises" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Nova Electronics" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Apex Traders" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Prime Works" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Core Depot" }, { "cityLocatedIn": "Kolkata", "distanceKm": 358, "supplierName": "Swift Parts" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Tech Parts" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Sigma Components" }, { "cityLocatedIn": "Visakhapatnam", "distanceKm": 403, "supplierName": "Star Solutions" }
```

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Product and Supplier ...

Top 5 Suppliers per City

City	Count
Surat	14
Gurugram	14
Coimbatore	14
Jaipur	14
Hyderabad	14

Top 10 Suppliers of Product

Supplier	Count
Hense Hub	110
Core Systems	90
Apex Tech	85
Cyber Tech	85
Prime Depot	85
Hense Elett	80
Nova Supply	75
Global Tech	70
Core Elett	65
Sigma Hub	65

No. of Suppli...

328

No. of Produc...

49

Top 5 Most Expensive Products

Product Name	Price
GPU NVIDIA RTX 3060	35882.65
CPU Intel Core i7-12700	30529.77
CPU AMD Ryzen 7 5800X	26737.0
Monitor 27" QHD	23598.7

Geographic Distribution of Cities with Suppliers